

Explainability-Guided SegFormer for Forest Cover Segmentation Using Attention Consistency Supervision

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Abstract

Vision Transformer (ViT)-based segmentation models achieve strong accuracy, but their internal attention is typically used only for post-hoc visualization, not as a supervised training signal. Prior work applying attention supervision for interpretability has focused on medical imaging and autonomous-driving domains, while forest/non-forest segmentation from aerial imagery still treats attention purely as an architectural feature-refinement mechanism, with interpretability evaluated qualitatively if at all. We present an explainability-guided SegFormer-B0 architecture that treats attention as a first-class training target: an *Attention Consistency Loss* aligns the model’s internal attention with the ground-truth forest mask during training, encouraging the network to focus on canopy regions rather than roads, shadows, or built structures. Because SegFormer’s encoder uses spatial-reduction attention rather than standard multi-head self-attention, we adapt Gradient-weighted Attention Rollout to SegFormer, restricting it to the final encoder stage where attention is square and token-consistent. We introduce a quantitative metric, Average Attention–Mask Overlap (AAMO), to evaluate interpretability numerically rather than only visually. In preliminary small-scale experiments, the proposed method raises AAMO from 0.0144 to 0.432 relative to a vanilla SegFormer-B0 baseline while also slightly improving Dice (0.8017→0.8191) and IoU (0.6690→0.6936), alongside a U-Net baseline (Dice 0.8492). Full-scale validation and a complete ablation study are ongoing.

1 Introduction

Semantic segmentation of forest cover from aerial imagery underpins ecological monitoring, deforestation tracking, and land-use management. Transformer-based models such as SegFormer [8] have been applied to this task for their ability to capture long-range spatial dependencies, but the great majority of transformer segmentation work, including recent forest-specific studies [1, 5, 7], uses attention purely as an architectural mechanism to improve accuracy, evaluating explainability only through post-hoc qualitative heatmap inspection, if at all.

Transformer attention in segmentation models is not explicitly guided during training and frequently attends to task-irrelevant regions — roads, shadows, buildings — rather than the class of interest, reducing both accuracy at ambiguous boundaries and the trustworthiness of the model’s explanations. No existing work addresses this specifically for forest/non-forest segmentation, nor evaluates attention faithfulness quantitatively in this domain.

This paper makes four contributions: (1) an Attention Consistency Loss that supervises SegFormer-B0’s internal attention

against the ground-truth forest mask during training; (2) an adaptation of Gradient-weighted Attention Rollout to SegFormer’s spatial-reduction attention; (3) AAMO, a quantitative interpretability metric evaluable alongside standard segmentation metrics; and (4) a preliminary empirical comparison against U-Net and vanilla SegFormer-B0 baselines showing that explainability-guided training does not cost — and may slightly improve — segmentation accuracy while substantially improving attention faithfulness.

2 Related Work

Existing work relevant to this proposal falls into two families.

Attention-as-architecture. ForResANeXt [5] embeds light-weight attention in residual blocks for forest/non-forest satellite segmentation; Attention-Refined PP-LiteSeg [7] fuses multi-branch attention for cross-regional UAV forest segmentation; an attention-based U-Net [1] adds attention gates for deforestation mapping on Sentinel-2 imagery. All three use attention purely as a feature-refinement mechanism improving accuracy — attention is never supervised or evaluated as an explanation.

Attention-as-explainability. eX-ViT [9] regularizes ViT attention with an attribute-guided loss for weakly supervised natural-image classification; TransAttUnet [4] integrates guided attention into a U-Net/Transformer hybrid for medical segmentation; work on driving models [6] applies an explicit attention loss to align attention with task-relevant scene regions. All three supervise attention as a training signal, but outside the forest/remote-sensing domain and without a quantitative attention-mask overlap metric. The attention-extraction method underlying our approach is Chefer et al.’s Gradient-weighted Attention Rollout [3], originally defined for standard multi-head self-attention.

Gap. No prior work combines (a) attention supervision as a training signal, (b) the forest/non-forest remote-sensing domain, and (c) a quantitative interpretability metric evaluated alongside standard segmentation metrics. This paper addresses that combined gap. We also draw on the WildUAV forest inspection dataset work [2] as evidence that forest-aerial-segmentation research so far has focused on data/architecture rather than explainability.

3 Proposed Method

3.1 Architecture Overview

The pipeline (Figure 1) extends SegFormer-B0 with an explainability-supervision path applied during training only; at inference the model runs as standard SegFormer-B0 with negligible overhead. The encoder’s stage-4 attention (the only stage with spatial-reduction ratio = 1, giving square, token-consistent 64×64 self-attention) feeds an adapted Gradient-weighted Attention Rollout to produce

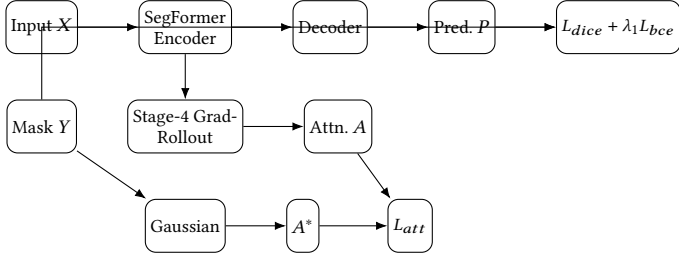


Figure 1: Training pipeline: predicted mask P and attention map A are each supervised, then combined into the total training objective L .

attention map A . The ground-truth mask Y is Gaussian-smoothed into a soft target A^* . A and A^* are compared by the Attention Consistency Loss and combined with the standard segmentation loss.

3.2 Handling Spatial-Reduction Attention

Standard Attention Rollout assumes full multi-head self-attention across all tokens. SegFormer’s Efficient Self-Attention reduces keys/values by a spatial-reduction ratio at earlier stages ($sr = 8, 4, 2$ for stages 1–3), making those stages’ attention matrices rectangular rather than square. Only stage 4 ($sr = 1$) has true square, token-consistent 64×64 attention — the only stage where Rollout’s recursive matrix product is valid without further approximation, verified empirically on MiT-B0 at 256×256 input resolution.

3.3 Mathematical Formulation

Let $X \in \mathbb{R}^{256 \times 256 \times 3}$ and ground-truth mask $Y \in \{0, 1\}^{256 \times 256}$:

$$F = E(X), \quad P = D(F) \in [0, 1]^{256 \times 256} \quad (1)$$

$$A = \text{Rollout}(F) \in [0, 1]^{256 \times 256} \quad (2)$$

$$A^* = \text{Gaussian}(Y) \quad (3)$$

$$L_{\text{dice}} = 1 - \frac{2|P \cap Y|}{|P| + |Y|}, \quad L_{\text{bce}} = - \sum [Y \log P + (1-Y) \log(1-P)] \quad (4)$$

$$L_{\text{att}} = \text{MSE}(A, A^*) \text{ or } \text{KL}(A \parallel A^*) \quad (5)$$

$$L = L_{\text{dice}} + \lambda_1 L_{\text{bce}} + \lambda_2 L_{\text{att}} \quad (6)$$

with $\lambda_1=1, \lambda_2=0.3$ initially, tuned on the validation set. Because A is itself defined via a gradient (Grad-Rollout is Grad-CAM-style), back-propagating L_{att} requires double backpropagation; our implementation keeps A attached to the autograd graph via `create_graph=True` so one combined `backward()` reaches all parameters.

4 Experimental Setup

Dataset. 5,108 RGB aerial images (256×256) with binary forest/non-forest masks; 70/15/15 stratified train/val/test split. A full dataset audit confirms 0 orphaned image/mask pairs, 0 unreadable files, and that masks are already effectively binary — no pixel sits more than 9 grey levels from pure black or white, and the mid-gray band (32–223) is empty. Preliminary results below use a 400/60/60 CPU-scale subset, not the full 5,108 images.

Table 1: Preliminary comparison (CPU/local smoke runs; SegFormer variants: 400/60/60 images, 8 epochs; DeepLabV3+: 200-image smoke run, optional extra baseline).

Model	Dice	IoU	F1	AAMO
U-Net (CNN)	0.8492	0.7379	0.8492	n/a
DeepLabV3+ (MobileNetV3)	0.7369	0.5834	0.7369	n/a
SegFormer-B0	0.8017	0.6690	0.8017	0.0144
SegFormer-B0 + L_{att}	0.8191	0.6936	0.8191	0.432

Baselines. U-Net (CNN) and vanilla SegFormer-B0 (no attention loss), isolating the effect of the proposed attention supervision.

Metrics. Dice, IoU, F1 (segmentation quality); parameter count, GFLOPs, FPS (efficiency); AAMO — normalized overlap between thresholded attention and the ground-truth forest mask (interpretability).

5 Preliminary Results

Table 1 reports an 8-epoch, 400-image CPU smoke run. AAMO increases approximately $30\times$ ($0.0144 \rightarrow 0.432$) with the attention-consistency variant, while Dice and IoU also improve slightly over vanilla SegFormer-B0, suggesting the attention supervision does not cost segmentation accuracy at this scale. We additionally report DeepLabV3+ (MobileNetV3 backbone) as an optional CNN reference point beyond U-Net; on a 200-image smoke run it trails both U-Net and SegFormer-B0 on Dice/IoU and shows a strong precision/recall imbalance (precision 0.59, recall 0.98), indicating a bias toward over-predicting forest rather than delineating boundaries precisely.

Limitations of these preliminary numbers. Eight epochs on 400 images is a smoke-scale proof that the pipeline is correct end-to-end, not a reliable trend; the full 5,108-image, 20-epoch run is pending. Qualitative attention-drift figures do not yet show as clean a story as the AAMO number: the vanilla model’s attention exhibits a corner artifact rather than the motivating roads/shadows failure mode, plausibly from undertraining, and warrants revisiting once the full-scale model has properly converged.

6 Discussion

The large AAMO gain relative to a small Dice/IoU change suggests attention faithfulness and segmentation accuracy are, at least partially, separable objectives at this scale — supervising attention does not appear to trade off against accuracy, though this must be confirmed with the full-scale run and a multi-seed ablation study before drawing firm conclusions. Both the MSE and KL-divergence formulations of L_{att} are implemented; only MSE has been evaluated so far.

7 Conclusion and Future Work

We presented an explainability-guided SegFormer-B0 that supervises internal attention against the ground-truth forest mask via an Attention Consistency Loss, an adaptation of Gradient-weighted Attention Rollout to SegFormer’s spatial-reduction attention, and a new quantitative interpretability metric, AAMO. Preliminary small-scale results support the core hypothesis. Future work will run

the complete multi-seed ablation study at full scale, compare the MSE and KL-divergence loss formulations, tune λ_2 via a validation sweep, and, time permitting, add a boundary-refinement module targeting canopy-edge errors.

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